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# Natural Computation

## Compact Genetic Algorithm- Benchmarks



Chumakov, Datzmann, Demir, Waclawek

# Easy Task - OneMax (Bit Counting)

**Goal :**The goal is to maximize the number of "1"s in a bitstring of length L.

**Utility:** This serves as a "sanity check." If your cGA cannot solve OneMax quickly, there is a fundamental bug in the implementation or parameter settings

$$f_{trap(k)}(x^{(i)}) = \begin{cases} k - u(x^{(i)}), & \text{if } u(x^{(i)}) < k \\ k, & \text{otherwise} \end{cases}$$

# Medium Task - Deceptive Trap Function

Bitstring of length L is divided into smal **m = L / k** non-overlapping blocks of **k bits each**.

Within each block, fitness is defined by trap function, where **u is number of 1s** in the block, then:

- **trap(t) = u** **if u = k**
- **trap(t) = k - u (- 1)** **if u < k**

Thus, fitness increases as the number of zeros increases, except for the case of all ones, which has the highest fitness

# Medium Task - Deceptive Trap Function - example

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **u** | 0 | | 1 | | 2 | | 3 | | 4 |
| **trap(**u**)** | | 4 | | 3 | | 2 | | 1 | | 4 |
| **trap(**u - 1**)** | | 3 | | 2 | | 1 | | 0 | | 4 |

“For each of the trap function, a hill climber is expected to be either randomly initialized at the global optimum, or to be at the slope leading to the local optimum. **Once trapped at the local optimum, the only way to get out is by performing a k bits jump, which is exponentially unlikely with growing k**.”

https://arxiv.org/pdf/2407.08335v1

# Medium Task - Deceptive Trap Function

**Total fitness** is defined as sum over all block with *trap function applied to each block* (from 1 to m).

$$f_{MK}(x) = \sum_{i=1}^{m} f_{trap(k)}(x^{(i)})$$

- tests the cGA's diversity and its ability to **avoid premature convergence**.
- highlights the impact of the "virtual population size" on escaping local optima: with smaller n, step size 1/n is smaller, thus **probabilities fix on 0 before the convergence shows that 1 is global optimum** …
- … more steps => higher chance of generating block of all 1s, where fitness = k, so that convergence starts in correct direction towards 1

# Hard Task - Circle Packing

**The Container:** A 2D unit square with boundaries at [0,1] for both x and y axes.
**The Objects:** N circles with a fixed radius r.
**Goal:** Arrange the circles such that they do not overlap with each other and do not cross the boundaries of the square.

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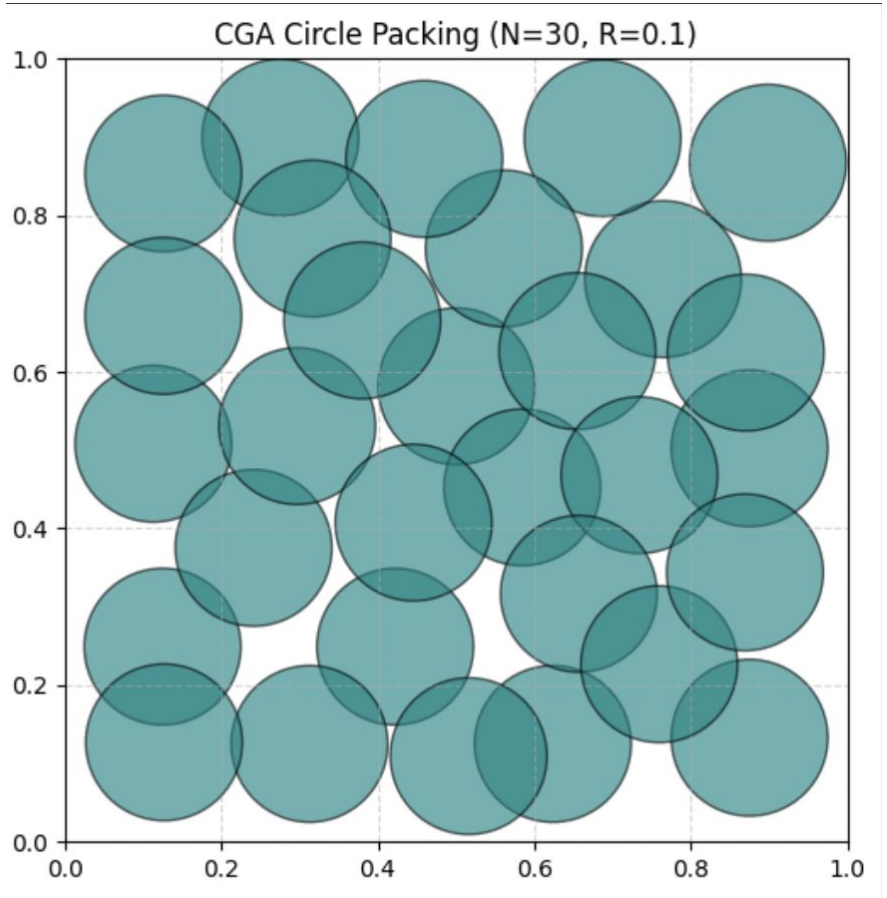

# Circle Packing - Modeling and Representation

**Genotype (Binary):** Each circle's x and y coordinates are encoded into bitstrings (default: 10 bits per value). For N circles, the total vector length is N×2×10.
**Phenotype (Decoding):** The bitstring is converted into a real number in the range [0,1] using the following mapping:

$$val_{float} = \frac{\sum_{i=0}^{k-1} b_i \cdot 2^{k-1-i}}{2^k - 1}$$

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# Circle Packing - Fitness Function Formula

We have 2 criteria :

- Boundary Violation
- Overlap Penalty

$$Fitness = \sum_{i=1}^{N} (P_{bound}(x_i) + P_{bound}(y_i)) + P_{overlap}$$

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# Circle Packing - Fitness Function Formula

$$P_{bound}(x) = \max(0, r - x) + \max(0, x + r - 1)$$

$$P_{bound}(y) = \max(0, r - y) + \max(0, y + r - 1)$$

$$P_{overlap} = \sum_{i<j} \max(0, 2r - d_{ij})^2$$